Solving Hidden Quadratics using Substitutions

Exam code: 9MA0

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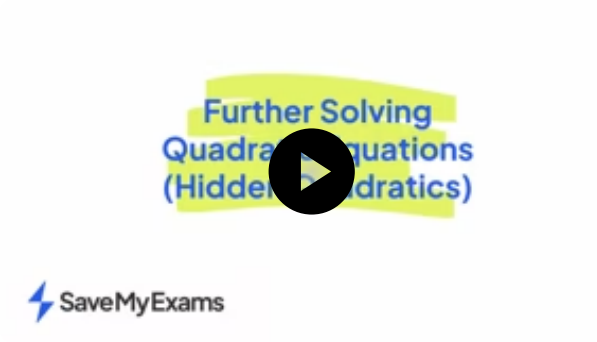
Updated on 19 June 2025



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Help me understand

Solving hidden quadratics using substitutions

What are hidden quadratic equations?

- Hidden quadratic equations are quadratics written in terms of a function $f(x)$
- A normal quadratic appears in the form $ax^2 + bx + c = 0$
- Whereas a hidden quadratic appears in the form $a[f(x)]^2 + b[f(x)] + c = 0$
- This might look complicated but it simply means x has been replaced by $f(x)$
- e.g. $\sin^2 x + 2\sin x - 3 = 0$ is just the hidden quadratic of $x^2 + 2x - 3 = 0$ where $f(x) = \sin x$

How do I solve hidden quadratic equations?

- First rearrange the function into the form $a[f(x)]^2 + b[f(x)] + c = 0$
- Replacing the function and solving the 'normal' quadratic first
- Then substitute the function back into the solutions to solve the original quadratic

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

$a[f(x)]^2 + bf(x) + c = 0$

$\cos^2 x + 2\cos x - 3 = 0$

STEP 2:
- "IGNORE" THE FUNCTION OF x AND SOLVE AS A NORMAL QUDRATIC
- USE A SUBSTITUTION, $y = f(x)$ IF YOU PREFER

$y = \cos x, \quad y^2 + 2y - 3 = 0$

$(y+3)(y-1) = 0$

$y = -3 \quad y = 1$

STEP 3: USE THESE SOLUTIONS, AND THE FUNCTION OF x, TO SOLVE THE ORIGINAL QUESTION

$\cancel{\cos x = -3}$ NO SOLUTIONS

$\cos x = 1$

$x = 0°, 360°$

Worked Example

Solve the equation $3^{2x+1} = 82 \times 3^x - 27$.

$3^{2x+1} - 82 \times 3^x + 27 = 0$

REARRANGE SO IT "= 0" AND WORK TOWARDS CORRECT FORM

$3^1 \times 3^{2x} - 82 \times 3^x + 27 = 0$

AWKWARD TO SPOT BUT 82×3^x IS THE CLUE

$3(3^x)^2 - 82(3^x) + 27 = 0$

THIS IS NOW IN THE FORM $a[f(x)]^2 + bf(x) + c = 0$

LET $y = 3^x$

$3y^2 - 82y + 27 = 0$

$(3y-1)(y-27) = 0$

$y = \frac{1}{3} \quad y = 27$

YOU DO NOT HAVE TO USE A SUBSTITUTION

$\therefore 3^x = \frac{1}{3} \quad 3^x = 27$

$x = -1 \quad x = 3$

MAKE SURE YOU FIND x's FOR YOUR FINAL ANSWER

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